DiarizBench

A diarization leaderboard for multilingual product teams

Voice Arena take-home · Q1 · diarization benchmark design and measured pilot

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Five working days · free tools · Kaggle T4 ×2 · experiment frozen before writing

This document designs a public diarization leaderboard for multilingual product teams and reports a fully executed pilot on 24 recordings, three open systems, and 72 inference runs. It separates two scopes throughout: the **proposed production benchmark** (a complete design) and the **measured pilot** (what was actually run and scored). Every number in the pilot traces to a saved artefact; the provenance table is Appendix C.

24	7	3	72	196
recordings	languages	systems	runs	min speech

In this submission. (1) This main report: design, measured pilot, analysis, and reflections. (2) A one-page summary for a decision-maker. (3) A leaderboard sketch with the overview table, model-detail view, and governance rules. (4) A two-minute video walkthrough. Code, manifest, environment pins, and all result files ship as reproducibility material.

How to read this document. The full argument is 14 sections plus three appendices. If you are short on time, the table below is the shortest honest route to a decision.

If you want to…	Read
Get the result in two minutes	Executive summary, then the one-page summary
Judge whether the design is right	Sections 1–4: audience, metrics, test data, frozen contract
Judge whether the evidence supports it	Sections 6–8: execution, results, significance testing
See where it breaks	Sections 9–12: three failure analyses, annotation audit, threats to validity
See what I would change	Sections 13–14: production plan, then the four reflections

Contents

Executive summary . . . 3
1. The product question . . . 4
2. What the leaderboard measures, and why . . . 4
2.1 Two scopes, kept separate . . . 4
2.2 Research that changed a decision . . . 5
2.3 Fairness rules (frozen before inference) . . . 5
3. Test data . . . 5
3.1 The 24 recordings . . . 5
3.2 Scoring regions and excerpt policy . . . 6
3.3 References and their conventions . . . 6
3.4 Proposed production collection (not collected) . . . 6
4. Metrics and scoring contract . . . 7
4.1 The metrics . . . 7
4.2 Frozen settings . . . 7
4.3 Validating the instrument before using it . . . 8
5. Systems under test . . . 8
5.1 The three pilots . . . 8
5.2 Two additional systems were attempted but not benchmarked . . . 8
5.3 A deviation that must be disclosed: Nemotron ran in streaming mode . . . 9
5.4 Broader comparison catalog: twelve named systems . . . 9
6. Execution and reproducibility . . . 10
6.1 Pipeline . . . 10
6.2 The environment was the hard part . . . 10
6.3 Timing protocol . . . 11
7. Results . . . 11
7.1 The main table . . . 11
7.2 Per corpus . . . 11
7.3 Indic languages . . . 12
7.4 Where the error comes from . . . 13
7.5 Speaker counting . . . 13
7.6 Low-activity speaker coverage . . . 14
7.7 Overlap . . . 14
7.8 Collar sensitivity . . . 15
7.9 Speed . . . 15
8. How sure are we that one system beats another? . . . 16
9. Three failure analyses . . . 17
9.1 A brief speaker disappears (BN-01, Bengali near-field) . . . 17
9.2 Three people talk at once (TE-02, Telugu in-the-wild) . . . 17
9.3 A multilingual slice separates tested configurations (TA-02 Tamil vs TE-02 Telugu) . . . 17
10. Annotation audit: what the references disagree about . . . 18
11. Did the research-inspired hypotheses hold? . . . 18
12. Threats to validity . . . 19
13. From pilot to production leaderboard . . . 20
13.1 Roadmap toward a speech evaluation lab (proposed) . . . 20
13.2 Extending metrics, datasets, and deployment evidence (proposed) . . . 21
References . . . 22
Appendix A. Reproducibility . . . 23
Appendix B. Per-recording results . . . 23
Appendix C. Number provenance . . . 24
14. Reflections . . . 25
14.1 Which design choice are you least sure about, and why? . . . 25
14.2 Which benchmark you read influenced you most, and how? . . . 25
14.3 What could make your leaderboard misleading, and how did you guard against it? . . . 25
14.4 What would you improve with more time? . . . 25

Executive summary

A product team choosing a diarization engine is not asking "which model has the lowest DER". It is asking whether the system knows **who spoke when**, whether it keeps the **quiet participant** who says twelve seconds in a five-minute meeting, and whether it returns answers **fast enough and under the conditions the product actually runs in**. This submission turns those three questions into a leaderboard design, then tests the design by running it end to end.

What was built. A frozen scoring contract (zero collar, overlap scored, unknown speaker count), a 24-recording manifest across three corpora and seven languages, a validated scoring harness (25 tests, cross-checked against `dscore`), a checkpointed Kaggle pipeline, and a paired-bootstrap analysis. Three open systems were run on all 24 recordings: **72/72 units completed with zero errors**.

What was measured (equal-weight over the 24 recordings; DER zero collar, overlap scored):

System	DER % (95% CI)	JER % (95% CI)	Exact count	Low-activity recall	RTF
`pyannote/speaker-diarization-community-1`	20.4 [15.5, 25.3]	38.6 [30.3, 47.2]	38%	45%	0.072
`nvidia/Nemotron-3-Diarization` (streaming)	21.6 [15.7, 27.6]	32.5 [23.4, 41.6]	**58%**	**61%**	0.079
`CIIL/Anuvani-Diarization`	**13.8 [10.0, 17.9]**	**29.9 [22.7, 37.4]**	33%	47%	**0.053**

The measured configuration result. Anuvani has lower DER than both tested configurations by more than the predeclared one-point practical threshold and with Bonferroni-corrected significance (+6.6 pp vs community-1, +7.8 pp vs Nemotron). The DER difference between community-1 and Nemotron is **not resolved by this sample** (−1.2 pp, 95% CI [−4.2, +2.0]). The JER view is weaker still: it separates Anuvani from community-1 but not from Nemotron (+2.7 pp, CI [−3.6, +9.0]). We report both and claim only what survives.

What the headline hides. All three systems under-count speakers (bias −0.58 to −1.17 speakers per recording) and all three drop **39–55% of low-activity speakers** — people who speak at most 5% of the meeting. On the hardest Telugu conversation, overlap-only DER is 82–100% for every system while whole-file DER is 38–42%. Nemotron's error is 79% *missed speech* (88% on AMI), a pattern consistent with conservative speech activation, but this pilot does not isolate its cause.

The three design decisions that changed the answer.

1. **Score strictly and say so.** Zero collar, overlap included, unknown speaker count. A 250 ms collar lowers every score by 2.8–3.6 pp without changing the ranking, so the strict number is defensible *and* the sensitivity is disclosed.
2. **Publish speaker coverage next to accuracy.** the community-1/Nemotron DER gap is unresolved; count accuracy differs by 20.8 pp and low-activity recall by 15.8 pp descriptively. A leaderboard that shows only DER would erase a product-relevant difference.
3. **Never pool corpora silently.** The three corpora label overlap at 0.1%, 4–20%, and 16–28% and time segments at 11–746 per file. Per-corpus cells come first; the cross-corpus average is labelled exploratory.

The limitation that matters most. Anuvani was fine-tuned on Indic DiarBench, one of the three pilot corpora, so its Indic margin cannot be separated from training-data exposure. Its wins on AMI and VoxConverse do not establish unseen-data generalisation: its DiariZen parent also lists these corpora as training sources. Exact session overlap has not been audited. The Indic comparison is in-domain, not unseen-language, generalisation. This is disclosed on every score, not buried.

1. The product question

Diarization leaderboards usually rank systems by diarization error rate (DER) on a handful of English meeting corpora. That ranking answers a research question. It does not answer the questions a product team asks before committing engineering time, a vendor contract, or a privacy review.

We framed the leaderboard around three questions, and made each one a first-class output rather than a footnote:

- **Accuracy — does it identify who spoke when?** DER against a fixed, strict scoring contract, with an interval rather than a point estimate.
- **Speaker coverage — does it preserve people who speak briefly?** Exact speaker-count accuracy, count bias, and a low-activity-speaker recall diagnostic, because duration-weighted DER is mathematically capable of staying low while a participant disappears.
- **Practicality — how fast, and under what conditions?** Real-time factor (RTF) on fixed hardware with the timed workload stated exactly, model-load time reported separately, and operating mode (offline vs streaming) treated as a filter rather than a footnote.

The leaderboard serves the following users and decisions:

User	Decision supported	Evidence to inspect
Meeting-product or ML lead	Which engine should power participant-attributed notes, and which needs a deeper trial?	DER/JER, brief-speaker coverage, overlap failures, and the audio inspector
Call-analytics buyer	Which service merits a vendor evaluation under telephone, language, privacy, and latency constraints?	Telephone/language cells, delivery mode, coverage, cost and workload disclosures; telephone testing is proposed, not present in this pilot
Multilingual voice-product builder	Does a general or Indic-adapted system suit the product's actual language mix?	Indic language/domain slices, code-mixing annotation limits, exposure and confidence intervals
Benchmark maintainer	Can this row be published, reproduced, compared, or re-scored after a correction?	Complete run evidence, protocol/reference versions, settings, failures and provenance

The pilot supports shortlisting and diagnostic questions. It cannot replace a product-specific, held-out trial, establish vendor privacy properties, or provide production cost/latency estimates.

> **What this document is not.** It is not a claim that three systems on 24 recordings settle the ranking of diarization. The pilot is deliberately small and deliberately strict about what it can support: it validates the *instrument*, demonstrates that the instrument changes conclusions, and publishes intervals wide enough to show how much is still unknown.

2. What the leaderboard measures, and why

2.1 Two scopes, kept separate

Scope	What it contains	Status in this submission
Proposed production benchmark	Collection and annotation protocol, unified reference convention, eight domain × language cells, governance and re-scoring rules	Designed, not built
Measured pilot	3 open systems × 24 recordings, frozen contract, scored and analysed	Executed end to end

Nothing in the production design is presented as measured. Where a production rule could not be implemented at pilot scale — the unified annotation convention, the eight-cell equal weighting, live-streaming latency — the pilot says what it substituted and why.

2.2 Research that changed a decision

Each source below earned its place by changing something concrete in the design, not by being cited.

Source	Finding that mattered	Decision it forced
Beyond DER (2026)	Duration-weighted DER hides dropped low-activity speakers	Mandatory count metrics + low-activity recall diagnostic
Indic DiarBench (2026)	Code-mixed, overlapping Indian conversation is where English-centric systems diverge	Dedicated Indic cells; 12 of 24 pilot recordings
SDBench / OpenBench (2025)	Accuracy without measured speed is not a product answer	RTF with a stated workload; fixed hardware; load reported separately
DIHARD III (2020)	Strict scoring (no collar, overlap scored, unknown count) is the honest default	Contract frozen before any inference
Anuvani-Diarization (2026)	Annotation conventions are worth 1.9–4.1 pp of DER	Convention audit; never pool corpora silently
Benchmarking Diarization Models (2025)	Aggregate DER hides *where* systems fail	Missed / false-alarm / confusion decomposition
DISPLACE 2024	Speaker changes and language changes are entangled in code-switched speech	Language slices; speaker-vs-language separation named as a production requirement
Park et al. review (2021)	Modular vs end-to-end families fail differently	System-family column; both families represented in the pilot

The single most consequential of these was DIHARD III. Freezing the contract before inference is what makes the 250 ms sensitivity result meaningful: it is a *pre-declared* alternative, not a post-hoc rescue of a number that looked bad.

2.3 Fairness rules (frozen before inference)

1. **Identical input.** Every system received the same 16 kHz mono waveform and the same scoring regions. No system received reference transcripts, identities, or a speaker-count hint.
2. **Frozen defaults.** Documented default inference settings, fixed before evaluation. No tuning on pilot audio.
3. **No reference-driven repair.** No gap filling, no speaker re-labelling against the reference, no system-specific post-processing beyond the model's own documented output format.
4. **Exposure disclosed.** Every score carries a training-data exposure flag: known overlap, declared disjoint, or unknown.
5. **Operating modes separated.** Offline and streaming results never share a ranking. This rule turned out to matter more than expected (Section 5.3).
6. **Failures recorded.** An errored or unsupported run is reported, never silently dropped; an incomplete system gets coverage information and no overall rank.

3. Test data

3.1 The 24 recordings

The manifest was frozen before any prediction was inspected. Each row carries source, split, language, condition, scoring region, reference path, audio checksum, and per-system exposure.

Corpus	Recordings	Languages	Speakers per excerpt	Segments per file	Scored audio	Reference speaker-time	Overlap share
Indic DiarBench	12	Hindi, Bengali, Marathi, Tamil, Telugu, Kannada (2 each)	3–9	23–163	144–300 s	54 min	0.1–28.3%
AMI (SDM)	6	English	4	169–746	839–2143 s	115 min	3.9–20.5%
VoxConverse	6	English	6–13	11–49	300 s	27 min	0.0–8.8%

In total: **205 minutes of scored audio and 196 minutes of reference speaker-time** (speaker-time exceeds wall-clock where speakers overlap). Coverage targets were set in advance and met:

- **Crowded conversations:** 17 of 24 excerpts contain five or more speakers inside the scoring region (18 by whole-recording counts). Target: at least 6.
- **Substantial overlap:** 15 of 24 excerpts have at least 5% overlap. Target: at least 6.
- **Language spread:** six Indian languages, two recordings each, one near-field and one in-the-wild.

The Indic in-the-wild files are the code-mixed ones: the corpus documents English code-mixing, dialectal variation, and frequent overlap, and its transcripts interleave Indic script with Roman-script English. That is the slice the leaderboard exists to expose.

3.2 Scoring regions and excerpt policy

AMI is scored as whole meetings (839–2143 s) because the reference is word-timed and truncating a meeting changes the speaker set more than it saves compute. Indic in-the-wild and VoxConverse use deterministic 300 s excerpts; Indic near-field files are shorter than 300 s and are used whole (144–206 s). Audio is truncated to the scoring region **before** inference, so no system pays for or sees audio outside the scored window.

Excerpts cost speakers. Tamil `TA-02` has six speakers across the full recording but three inside the excerpt; Kannada `KN-02` drops 8 to 7; VoxConverse `VC-01` drops 15 to 13. All counts in this document are **in-region** counts, per the frozen contract. An earlier draft used whole-recording counts and was corrected; that correction alone moved community-1's count MAE from 1.54 to 1.17.

3.3 References and their conventions

The pilot uses each corpus's native reference rather than re-annotating: Indic DiarBench's speaker-attributed transcripts converted to RTTM (12/12 validated against the dataset's own speaker counts), AMI's official `only_words` RTTM, and VoxConverse's official `dev` RTTM. This is the honest choice for a five-day pilot and the wrong choice for a production benchmark, because the three conventions disagree (Section 10). The production design replaces them with one unified convention; the pilot publishes per-corpus results first and labels every cross-corpus aggregate exploratory.

A conversion bug found at this stage is worth recording: in-the-wild samples label speakers as `Speaker A` and `New speaker A`. Spaces in the speaker field silently collapsed eight speakers into two when written as RTTM. Sanitising the field (`Speaker_A`) restored exact agreement with the dataset's declared counts on all 12 Indic files. Reference plumbing is where silent scoring errors live.

3.4 Proposed production collection (not collected)

The proposed benchmark contains **60 hours: 12 public development hours and 48 private evaluation hours**. Four domains each receive 15 hours: meetings, telephone, interviews/media, and informal conversation. Each domain is half English and half Indic; Hindi, Bengali, Marathi, Tamil, Telugu, and Kannada receive five hours each across the complete collection. At least 25% of evaluation recordings and duration should contain five or more speakers.

Target 2-party calls, 2–8-person meetings/interviews and dedicated 6–12-speaker crowded sessions. Collect headset/USB-mic, laptop/phone, tabletop/distant-mic and native telephone audio; keep native sample-rate metadata (upsampling an 8 kHz call does not restore its lost bandwidth). Balance regional accents and urban/rural speaker backgrounds within languages where recruitment allows, record self-declared metadata with consent, and publish representation gaps.

Record microphone/device, noise, accent/region (consent permitting), speaker count, turn length, overlap, and code-mixing. Use source-conversation and speaker-group-disjoint splits, keeping all microphone variants of a session together. Development material must permit the intended use; private evaluation conversations require explicit collection and evaluation consent.

Two independent annotators mark acoustic speech boundaries, brief verbal backchannels, and each simultaneous speaker separately. Pauses are excluded and non-speech vocalizations tagged separately. Measure pre-adjudication DER, count agreement, and boundary disagreement, then adjudicate with a third annotator and

version the references. The handbook and achieved agreement accompany scores. This is a collection and quality-control proposal, not an annotation study completed in this pilot.

4. Metrics and scoring contract

4.1 The metrics

Metric	Role	Definition
DER	Primary accuracy	(missed + false alarm + confusion) ÷ total reference speaker-time
JER	Companion accuracy	Mean matched-speaker Jaccard error: 1 minus intersection / union; unmatched reference speakers score 100%
Exact count accuracy	Speaker coverage	Recordings where predicted count equals reference count
Count MAE / bias	Coverage, magnitude and direction	mean abs error / mean signed error, in speakers
Low-activity recall	Proposed diagnostic	Share of low-activity speakers recovered after optimal one-to-one matching
Overlap-only DER	Diagnostic	DER restricted to reference time with two or more simultaneous speakers
DER components	Diagnosis	Missed, false-alarm, confusion, each as a share of reference time
RTF	Practicality	Processing time ÷ audio duration, with the timed workload stated

DER is duration-weighted, which is exactly why it is insufficient alone: a speaker who contributes 4.6% of a meeting can vanish entirely and move DER by a couple of points. JER gives each reference speaker equal weight and is the cheapest correction. The **low-activity diagnostic is our proposal**, not an established community metric, and is labelled as such wherever it appears.

Low-activity speaker: at most 5% of reference speaker-time in the recording, and at least 0.5 s of speech (a stability threshold for this diagnostic, not evidence that shorter speech is noise). **Recovered:** at least 50% of that speaker's reference time overlaps their optimally matched hypothesis speaker.

4.2 Frozen settings

Setting	Value	Why
Collar	**0 ms** headline; 250 ms sensitivity alongside	Strict scoring; the sensitivity shows how much is boundary jitter
Overlap	**Included**	Ignoring overlap flatters systems that cannot handle it
Speaker count	**Unknown** to every system	The system must estimate the count, which is the product requirement
Scoring regions	Explicit per recording	Prevents disagreement about lead-in, silence, and trailing audio
Count source	Labels with positive activity **inside** the region	A speaker who only speaks outside the excerpt is not a speaker for that recording
Output mode	Overlap-capable output preserved	Score the system as users receive it, not a mode tuned for the benchmark

Collar convention matters. A collar excludes a window around reference speech boundaries so small timing offsets receive less weight. In this harness, `pyannote.metrics collar=0.25` means a 250 ms total window, or 125 ms on each side, as implemented by `UEMSupportMixin.extrude`. The saved 250 ms sensitivity results use that exact setting; they must not be silently interpreted as a plus/minus 250 ms window from a different scorer. A collar can also hide brief-turn errors, so zero collar remains the primary result and overlap/speaker-count rules stay unchanged.

The headline number is therefore DER at zero collar, overlap scored, unknown count. Every other number is a companion or a sensitivity, never a substitute.

4.3 Validating the instrument before using it

Scoring was validated on nine hand-computed fixtures before any real audio was scored: perfect output (0.0% DER), permuted labels (0.0%), missing brief speaker (2.44% DER, 50% JER, count −1, low-activity recall 0), merged speakers (50% confusion), split speaker (50% confusion), overlap-perfect (0.0%), silence false alarm (50%), empty hypothesis (100%), and invalid timestamps (rejected). **25/25 tests pass.** DER and JER agree with `nryant/dscore` within 2% and 3% respectively on every fixture, after normalising dscore's percent units against pyannote's fractions.

One caveat is carried forward rather than hidden: Anuvani's model card reports that `pyannote.metrics` reads about 0.20 percentage points higher than NIST `md-eval` on Indic DiarBench. Our absolute numbers are therefore not directly comparable to md-eval numbers published elsewhere. Within-run comparisons use the same scorer consistently; this does not prove a universal constant offset.

5. Systems under test

5.1 The three pilots

System	Family	Speaker / overlap behaviour	Licence	Exposure (pilot corpora)	Settings
pyannote community-1	Modular: segmentation + embedding + VBx	No hard cap; overlap-capable	Open weights, gated access	**Known overlap** on AMI + VoxConverse; unknown on Indic	Unknown count; regular (non-exclusive) output
NVIDIA Nemotron-3	End-to-end Sortformer, ~100M parameters	**Hard cap 8 speakers**; overlap-capable	OpenMDW v1.1	Unknown	Streaming, low-latency mode (see 5.3); 8 channels
CIIL Anuvani	End-to-end (DiariZen `md-v2` + Indic fine-tune)	22 Indic languages + English	CC BY-NC 4.0 weights	**Known corpus exposure** on Indic DiarBench; parent training includes AMI + VoxConverse	Default; unknown count

Exact repository IDs: `pyannote/speaker-diarization-community-1`, `nvidia/Nemotron-3-Diarization`, `CIIL/Anuvani-Diarization`. These are the identifiers a reviewer can resolve; no family name stands in for a checkpoint.

The three were chosen to cover the families a product team will actually shortlist: a modular general baseline, a streaming end-to-end model, and an Indic-specialised end-to-end model. Published repository IDs and dependency constraints are recorded in `pins.md`. The run templates do not consistently pin immutable checkpoint revisions or capture full environment locks; exact future inference reproduction is therefore limited. Saved RTTMs make this scoring analysis repeatable.

5.2 Two additional systems were attempted but not benchmarked

- **DiariZen** `md-v2` (the parent of Anuvani) is blocked by its environment: Python 3.10, torch 2.1.1, and a vendored pyannote 3.1, with no conda on Kaggle. Its architecture is represented via Anuvani, its Indic derivative.
- **3D-Speaker CAM++** installed and ran but collapsed a six-speaker conversation to one speaker. The adapter was not validated further within the budget. This is an unsuccessful integration attempt, not evidence that the model is inferior. Excluding it after a poor smoke result also creates selection bias; a production benchmark must publish attempted integrations and failures.

The plan's fallback rule allowed exactly this: three working systems on the same 24 recordings, with the reduction disclosed. No recording was dropped.

5.3 A deviation that must be disclosed: Nemotron ran in streaming mode

The plan called for Nemotron's documented offline configuration. That was not achievable in this environment: the checkpoint's position limit is 5,000 frames (about 50 s), which makes a single forward pass over a 25-minute meeting unsafe, and NeMo's offline long-audio wrapper was not available. Nemotron was therefore run in the processor's **streaming, low-latency** mode, chunk by chunk, on every file.

This is an operating-mode mismatch and it is not cosmetic. Three observations motivate a matched-mode follow-up: Nemotron's error is **79% missed speech** overall and **88% on AMI**, a pattern consistent with conservative speech activation, although the cause was not isolated; its AMI DER (29.6%) is the highest of the three tested configurations; and Anuvani's card lists Nemotron at **18.38%** on Indic DiarBench under offline scoring, against our 25.1% Indic result. The honest reading is that **Nemotron's row is a streaming data point, not a verdict on its offline capability**, and the leaderboard must show operating mode as a column. We keep the run, label it, and retain its descriptive results without awarding it an offline rank. A same-data rerun in both modes is required to measure a mode effect; a published full-corpus score cannot estimate it.

5.4 Broader comparison catalog: twelve named systems

The assignment requires at least ten concrete candidates. Only the three pilots above have measured results. The following catalog covers self-hosted, cloud, and on-device products; it is not a 12-system evaluation. Public identifiers are given where available. Opaque API backends require the request configuration, response version metadata, region, and run date to be archived at execution. Documentation was reviewed on 11 November 2026; an API feature name is not a pinned checkpoint.

#	System / published identifier	Organization / family	Delivery and evaluation status
1	`pyannote/speaker-diarization-community-1`	pyannote; modular clustering	Self-hosted; measured offline
2	`nvidia/Nemotron-3-Diarization`	NVIDIA; end-to-end Sortformer	Self-hosted; measured streaming
3	`CIIL/Anuvani-Diarization`	CIIL; DiariZen-derived	Self-hosted; measured offline
4	`BUT-FIT/diarizen-wavlm-large-s80-md-v2`	BUT-FIT; WavLM-based diarization	Self-hosted; environment blocked
5	Precision-2 (`model="precision-2"`)	pyannoteAI; diarization pipeline	Cloud API; not run
6	Deepgram Nova-3 with versioned diarization (`diarize_model=v2`, batch)	Deepgram; ASR + diarization	Cloud API; not run
7	AssemblyAI Speaker Diarization (`speaker_labels=true`)	AssemblyAI; ASR + speaker labels	Cloud API; backend version/config to archive; not run
8	Amazon Transcribe (`ShowSpeakerLabels`)	AWS; ASR + speaker partitioning	Cloud API; backend not publicly pinned; not run
9	Speech-to-Text V2 Chirp 3 (`chirp_3`)	Google; ASR + diarization	Cloud API; not run
10	Speechmatics Batch Transcription with speaker diarization	Speechmatics; ASR + speaker labels	Cloud API; backend/config to archive; not run
11	Falcon (`pvfalcon`)	Picovoice; local diarization engine	On-device SDK; not run
12	SpeakerKit (`SpeakerKit`, Argmax OSS Swift)	Argmax; Core ML diarization	Apple-device SDK; not run; ASR optional

Commercial API systems require paid access, service-specific language checks, and a disclosed ASR workload where coupled. Self-hosted weights require their own license review: Anuvani and DiariZen weights are non-commercial. Speaker limits, overlap representation, supported languages, and immutable revisions must be checked for the exact execution configuration rather than inferred from a family name. Count limits above a model's capacity are retained and flagged, not excluded.

API elapsed time includes upload, queuing, polling, inference, and result retrieval. SDK speed is measured on the target device. Neither is directly ranked against Kaggle GPU inference-only RTF. Offline and streaming accuracy ranks are also separate; all-mode views have no shared rank.

Official product references: pyannoteAI, Deepgram, AssemblyAI, AWS, Google Chirp 3, Speechmatics, Falcon, and SpeakerKit.

6. Execution and reproducibility

6.1 Pipeline

The production evaluation proceeds through these eight steps; the pilot implements the saved-audio, inference, scoring and analysis stages, while consented collection and independent publication are proposals.

1. **Collect and register:** obtain permitted audio, consent where required, source IDs and metadata.
2. **Build references:** dual annotation, pre-adjudication agreement, adjudication and reference versioning.
3. **Freeze the test:** development/evaluation splits, scoring regions, 16 kHz mono conversion, checksums, collar/overlap/count policy and supported operating tracks.
4. **Register each system:** exact available identifier, immutable revision when obtainable, delivery mode, license/access, settings and training-exposure declaration. Missing version provenance stays visible.
5. **Run and preserve:** execute on the canonical audio, checkpoint recording-level status, and retain raw outputs, RTTMs, timing scope, errors and coverage without hiding unsupported cases.
6. **Score consistently:** validate RTTMs, apply the common UEM and scorer, compute accuracy, counting, overlap and timing diagnostics, and keep incomplete runs unranked.
7. **Assess and inspect:** paired conversation-level bootstrap, corrected comparisons and practical thresholds; inspect corpus/language slices and listen to selected examples without claiming a formal audit.
8. **Publish and maintain:** review the evidence, publish scores with uncertainty, mode/settings/hardware, coverage/exposure/reference versions and examples, archive prior boards, and re-score every affected system after a reference correction. API/device timing and offline/streaming ranks stay separate.

`manifest › inference › hypothesis RTTM › scoring › tables and figures`. Every unit (one system, one recording) writes its own RTTM, timing JSON, and status entry, and the driver skips a unit only when status is `ok` **and** the RTTM is non-empty. Re-running the notebook after an interruption resumes without repeating completed work. Raw output is zipped on Kaggle and merged locally by `scripts/t09/merge_output.py`, which applies the same done-rule as the orchestrator.

All 72 units completed on a Kaggle Tesla T4 ×2 in one notebook session, with **zero errors** and no missing recordings. Total inference time across the three systems was about 42 minutes of GPU time, well inside the 12 GPU-hour pilot budget.

6.2 The environment was the hard part

Three failures cost real time and are recorded because they are the difference between a reproducible run and a lucky one:

- **Broken mixed NumPy.** The image ships Debian NumPy 1.26.4 plus a partial pip 2.1.3, producing `numpy.ufunc has no __qualname__`. Resolution: `numpy>=2.5,<2.6` together with `numba>=0.67` (the first numba that accepts NumPy 2.5), plus an end-of-install forced NumPy reinstall as a safety net. Two earlier attempts (`numpy<2.3`; `numba>=0.60`) both left a broken ufunc state.
- **Torch silently upgraded.** Installing `pyannote.audio` pulled torch 2.15.0, which broke the runtime. Pinned `torch<2.15` and `torchvision<0.27` in the same install.
- **MP3 truncation.** `torchcodec` mis-read an MP3 as 9.93 s instead of 97.6 s. Every input is now 16 kHz mono WAV.

6.3 Timing protocol

Timing is measured on a fixed six-recording panel spanning 3–25 minutes, with one warm-up and three timed repetitions per recording. **Setup/load is timed separately and excluded from RTF.** The kernel times each system's `infer(wav)` callable, not an identical abstract model operation. File loading and required processing inside those callables are included. In particular, the Nemotron timing path returns frame logits without the pilot's final speaker-segment extraction; the other callables return pipeline outputs. These are measured implementation timings, not fully matched end-to-end workloads. No scoring or result upload is timed. Setup may include dependency checks/model initialization; Anuvani also performs its startup self-test. Three repetitions do not measure variation across fresh sessions or hardware. RTF is reported instead of "latency" because offline throughput says nothing about streaming responsiveness. Native streaming latency needs a live-input protocol, which is out of pilot scope and stated as such.

7. Results

7.1 The main table

Equal-weight over the 24 recordings. Intervals: paired bootstrap, 10,000 resamples, seed 20261111 for DER and JER; Wilson for count accuracy; recording-clustered bootstrap for low-activity recall (speakers are nested in recordings, so a naive interval would be too narrow).

System	DER %	JER %	Exact count	Count bias	Low-activity recall	DER @250 ms	RTF
community-1	20.4 [15.5, 25.3]	38.6 [30.3, 47.2]	38% [21, 57]	–1.17	45% [12, 69]	16.7	0.072
Nemotron-3	21.6 [15.7, 27.6]	32.5 [23.4, 41.6]	**58% [39, 76]**	–0.92	**61% [35, 81]**	18.8	0.079
Anuvani	**13.8 [10.0, 17.9]**	**29.9 [22.7, 37.4]**	33% [18, 53]	**–0.58**	47% [14, 72]	**10.3**	**0.053**

Table 1. Pilot leaderboard row per system. Exact repository IDs are given in Section 5.1. Counts are in-region. Low-activity recall is over 38 low-activity speakers across 24 recordings. Count MAE is in Table 3; model load time and RTF detail are in Table 6. RTF is the duration-weighted panel mean, load excluded. Completion coverage is 24/24 for every system.

The intervals are the story as much as the means: all three overlap, so the table alone cannot tell you whether Anuvani is genuinely ahead. That is why Section 8 tests the paired differences directly rather than asking a reader to eyeball overlapping intervals.

7.2 Per corpus

System	Indic (12)	AMI-SDM (6)	VoxConverse (6)	Corpus-equal
community-1	26.1	22.7	6.8	18.5
Nemotron-3	25.1	29.6	6.5	20.4
Anuvani	**16.5**	**17.5**	**4.9**	**12.9**

Table 2. DER % by corpus, native reference conventions. The corpus-equal column weights the three corpora equally; the global 24-recording mean is 20.4 / 21.6 / 13.8.

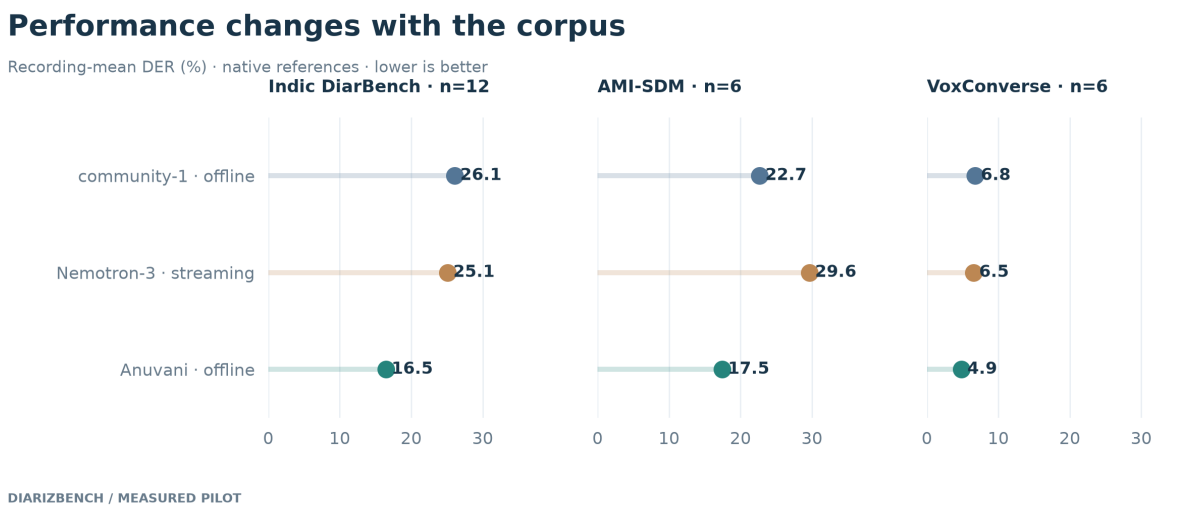


Figure 1. Per-corpus DER. Observed DER is lower on these VoxConverse excerpts than on the other corpora. Content, speaker participation and reference conventions all vary; this plot does not isolate their effects. Cross-corpus aggregates are exploratory.

Anuvani has the lowest observed mean in each corpus; the ordering of these tested means does not change under the reported weighting summaries. This does not remove exposure or mode confounds. The *margin* changes with weighting. On Indic it is 8.6–9.6 pp, on AMI 5.2–12.1 pp, on VoxConverse 1.6–1.9 pp. A leaderboard that ranked on VoxConverse alone would call the three systems nearly equal (4.9–6.8%); a leaderboard that ranked on Indic alone would call the gap huge. Neither is wrong; a single pooled number would hide both.

7.3 Indic languages

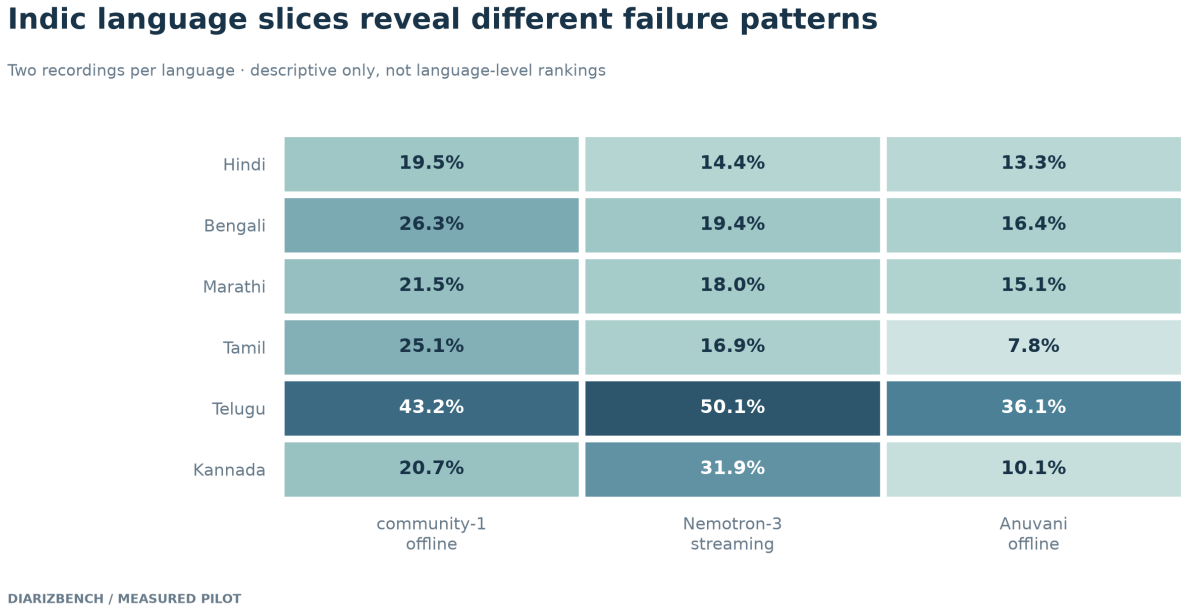


Figure 2. Indic DER by language, two recordings per language. Telugu floors every system; Tamil separates them most; Kannada reverses the community-1/Nemotron order. With n=2 per cell these are exploratory slices, not language rankings.

Telugu is the hard slice for everyone (36.1–50.1%) and it is the highest-overlap language in the pilot (20.4–28.3% on the two files). Tamil separates the systems most sharply: Anuvani 7.8% against 16.9% (Nemotron) and 25.1% (community-1). Kannada shows why two recordings per language cannot support a ranking: Nemotron's 31.9% is its worst Indic cell, while Anuvani's 10.1% is its best.

7.4 Where the error comes from

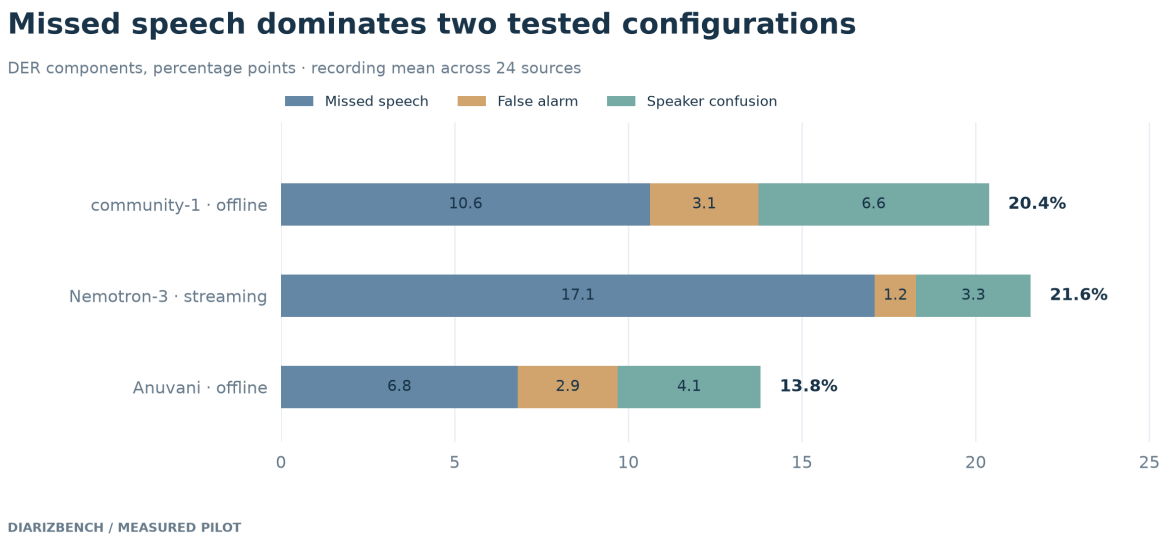


Figure 3. DER decomposed into missed speech, false alarm, and confusion. community-1 and Nemotron lose most of their error to missed speech; Anuvani is balanced and low on all three. Components sum to the DER in Table 1.

The decomposition changes the interpretation of the ranking. community-1's 20.4% is 10.6 missed + 3.1 false alarm + 6.6 confusion. Nemotron's 21.6% is 17.1 missed + 1.2 false alarm + 3.3 confusion: most of its error is missed speech. Configuration, model behavior, and reference conventions could all contribute; their effects were not isolated. Anuvani's 13.8% is 6.8 + 2.9 + 4.1, the only balanced profile. On Tamil `TA-02`, community-1's 29.6% is dominated by **confusion (22.1 pp)** with only 3.2 pp missed: it hears the speech and assigns it to the wrong person, collapsing three reference speakers into two.

7.5 Speaker counting

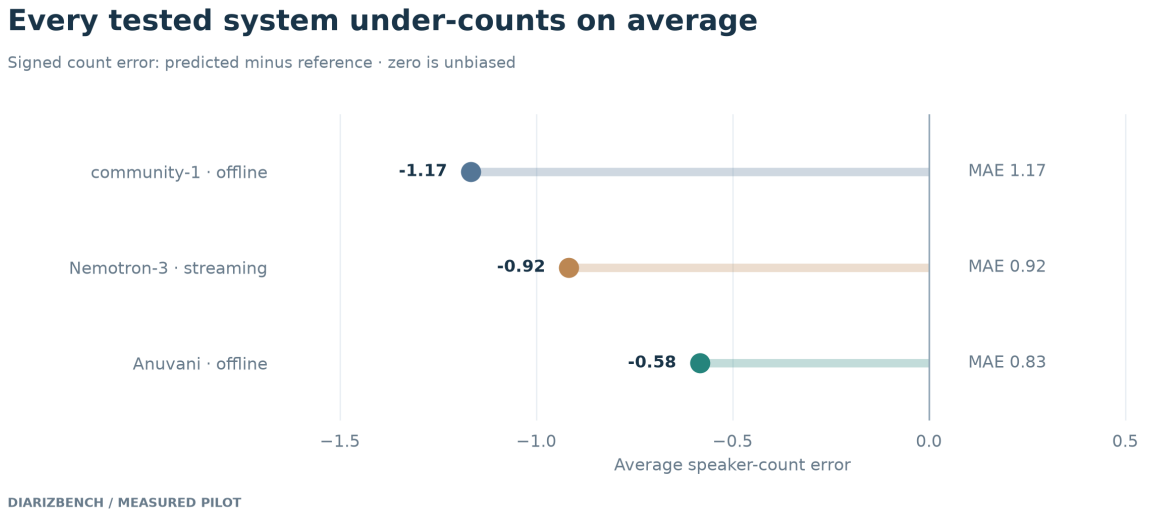


Figure 4. Signed speaker-count bias with mean absolute error. Every system under-counts on average; community-1 is the most biased, Anuvani the least.

System	Exact	MAE	Bias	Indic MAE	AMI MAE	VoxConverse MAE
community-1	9/24	1.17	–1.17	1.83	0.50	0.50
Nemotron-3	**14/24**	0.92	–0.92	1.42	**0.00**	0.83
Anuvani	8/24	**0.83**	**–0.58**	**1.08**	0.67	0.50

Table 3. Speaker-count behaviour, in-region counts.

Two different failure shapes hide behind similar MAE. Nemotron is exactly right most often (14/24, mostly the four-speaker AMI meetings) but carries the worst single error in the run: on `VC-01` it emits 8 speakers against 13 in the excerpt (15 in the full recording), a −5 error caused by its hard cap. Anuvani is rarely exactly right but rarely far off — a scattering of ±1 misses. community-1 systematically under-counts Indic (MAE 1.83, bias −1.83), which is the "beyond DER" failure the diagnostic exists to catch: on `BN-01` it finds 4 speakers where the reference has 8.

7.6 Low-activity speaker coverage

System	Recovered	Recall	95% CI	Recall at 3% / 10% activity
community-1	17/38	45%	[12, 69]	38% / 53%
Nemotron-3	23/38	61%	[35, 81]	58% / 62%
Anuvani	18/38	47%	[14, 72]	42% / 53%

Table 4. Low-activity recall at the headline 5% activity / 0.5 s duration / 50% recovery thresholds, with the contract-mandated sensitivity at 3% and 10% activity. The full 3×3 grid is in Appendix C.

The ordering is stable across the sensitivity grid — Nemotron first, community-1 and Anuvani indistinguishable — but the *level* is not: recall rises with the activity threshold because more speakers qualify and the easy ones are recovered. The intervals are wide (35–81% for the leader) because 38 speakers nested in 24 recordings is a small sample. **No system recovers even two thirds of the people who speak least**, and the best system on this diagnostic is the worst on DER. That combination is the argument for publishing coverage next to accuracy.

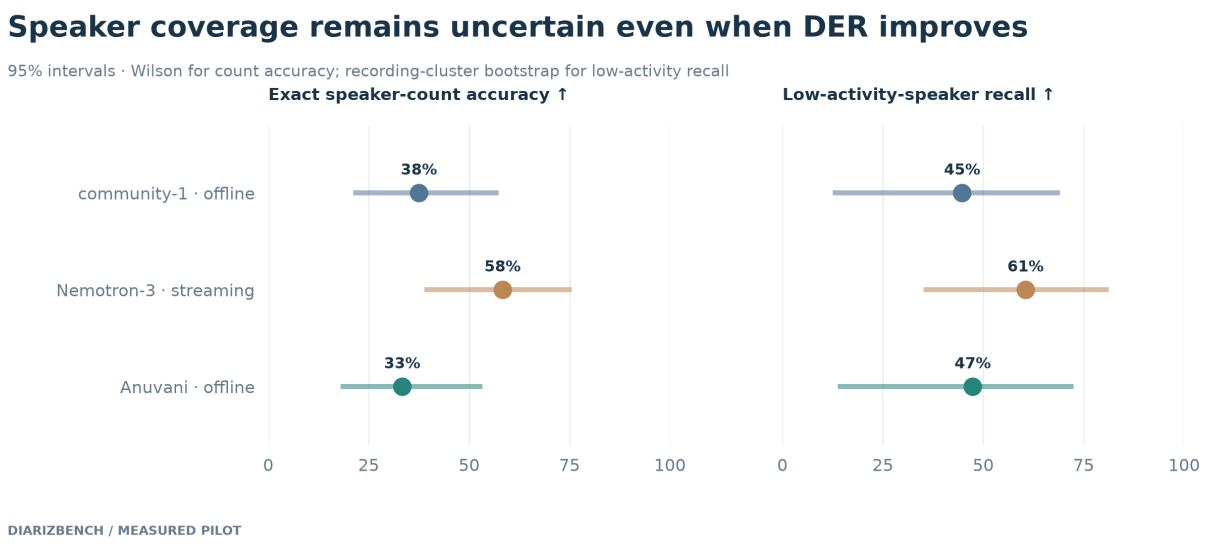


Figure 6. Exact count accuracy and low-activity recall with 95% intervals. Count intervals use Wilson; recall intervals resample recordings. Differences are descriptive and do not establish coverage superiority.

7.7 Overlap

Overlap is where the pilot's most uncomfortable number lives. Mean overlap-only DER across the 23 recordings that contain overlap is **68.2% (community-1), 59.9% (Nemotron), 67.7% (Anuvani)** — against whole-file DER of 20.4 / 21.6 / 13.8. On the densest moment of Telugu `TE-02` (three simultaneous speakers at 97.2–97.7 s) overlap-only DER is 99.8 / 81.8 / 91.7%.

Overlap-only DER does **not** track headline DER here: Anuvani has lower overall DER but worse overlap-only DER than Nemotron. community-1 has slightly higher overlap-only DER than Anuvani (68.2% versus 67.7%). We report the observation and resist the tidy explanation. At n=24 with 23 overlap-bearing files, the overlap view is a diagnostic that shows how much error concentrates in simultaneous speech, not a ranking signal. It does establish one thing clearly: a benchmark that skipped overlap would be measuring a different, easier task.

7.8 Collar sensitivity

System	DER, zero collar	DER, 250 ms collar	Change
community-1	20.4	16.7	–3.6
Nemotron-3	21.6	18.8	–2.8
Anuvani	13.8	10.3	–3.5

Table 5. Boundary-collar sensitivity. The 250 ms collar is a companion result, never a substitute for the zero-collar headline.

Removing a 250 ms boundary collar changes scores by about three points; this is boundary sensitivity, not proof that all removed errors are harmless jitter. The ranking is unchanged, which is the useful part: the conclusion does not depend on the collar, but the magnitude does, and a submission that reported only the 250 ms number would overstate every system by roughly 15% of its error.

7.9 Speed

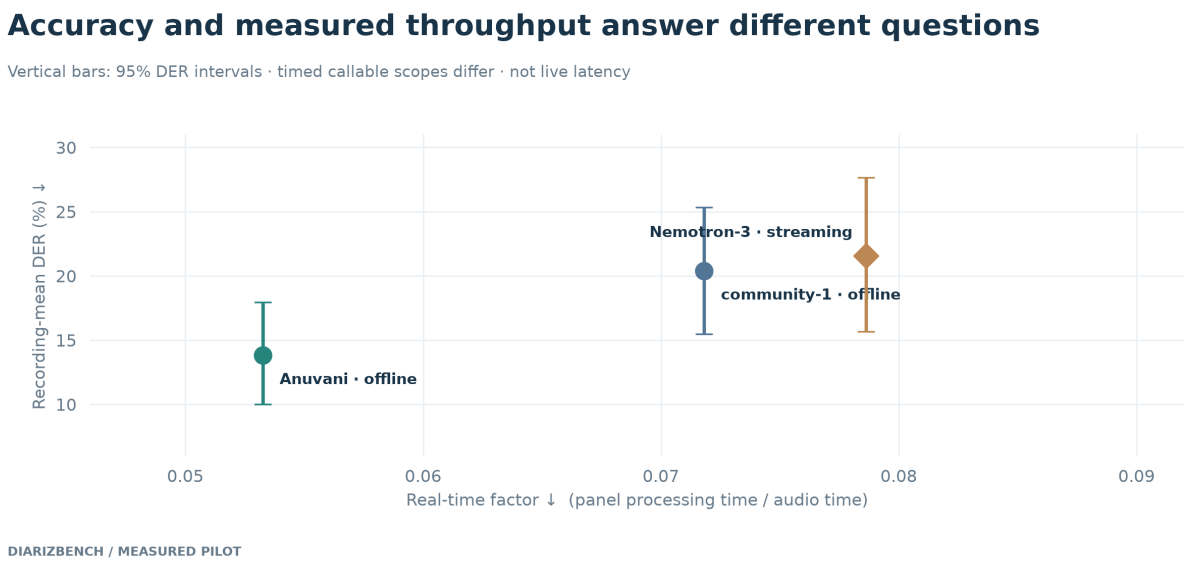


Figure 5. Accuracy against speed. Anuvani is bottom-left: most accurate and lowest RTF. Nemotron is top-right. Axes are DER (lower better) and RTF (lower better).

System	Setup/load (s)	RTF	Speed factor	RTF range across files
community-1	0.57	0.072	13.9×	0.064 (5 min) to 0.077 (25 min)
Nemotron-3	0.73	0.079	12.7×	0.078–0.079 (flat)
Anuvani	12.98	**0.053**	**18.8×**	0.051–0.054 (flat)

Table 6. Timing panel: one warm-up plus three repetitions on six recordings, one T4 ×2. Load excluded from RTF.

Two structural findings matter more than the ranking. First, **the modular pipeline degrades with length** (0.064 to 0.077) while the two end-to-end systems are flat, so RTF measured only on 5-minute clips would flatter community-1. Second, **load time inverts the ranking for short files**. For a single 5-minute recording, a setup + 300 × panel-RTF projection is 22.1 s (community-1), 24.3 s (Nemotron), 29.0 s (Anuvani) — community-1 is fastest. The break-even is about 11 minutes of audio; beyond that Anuvani has the lower projected total. These are extrapolations, not measured five-minute end-to-end service requests, and depend on whether setup is repeated. The unmatched timed-callable scope also prevents a universal speed claim.

8. How sure are we that one system beats another?

Paired bootstrap over the 24 independent recordings, 10,000 resamples, fixed seed. A win requires both statistical significance after Bonferroni correction across the three pairwise comparisons and the predeclared **≥1 pp DER** practical threshold.

Pair	Mean Δ DER (pp)	95% CI	Bonferroni CI	Verdict
community-1 vs Nemotron	–1.2	[–4.2, +2.0]	[–5.0, +2.6]	**not resolved**
community-1 vs Anuvani	+6.6	[+4.4, +9.1]	[+3.9, +9.7]	**Anuvani wins**
Nemotron vs Anuvani	+7.8	[+4.6, +11.1]	[+4.0, +12.0]	Mode-confounded

Table 7. Pairwise DER differences. Negative means the first system has lower DER. Mixed-mode rows describe tested configurations and do not establish model superiority.

Pair	Mean Δ JER (pp)	95% CI	Bonferroni CI	Verdict
community-1 vs Nemotron	+6.1	[–1.0, +12.6]	[–2.4, +14.2]	not established
community-1 vs Anuvani	+8.7	[+5.3, +13.1]	[+4.8, +14.3]	**Anuvani wins**
Nemotron vs Anuvani	+2.7	[–3.6, +9.0]	[–4.7, +10.2]	not established

Table 8. Pairwise JER differences, same resampling scheme.

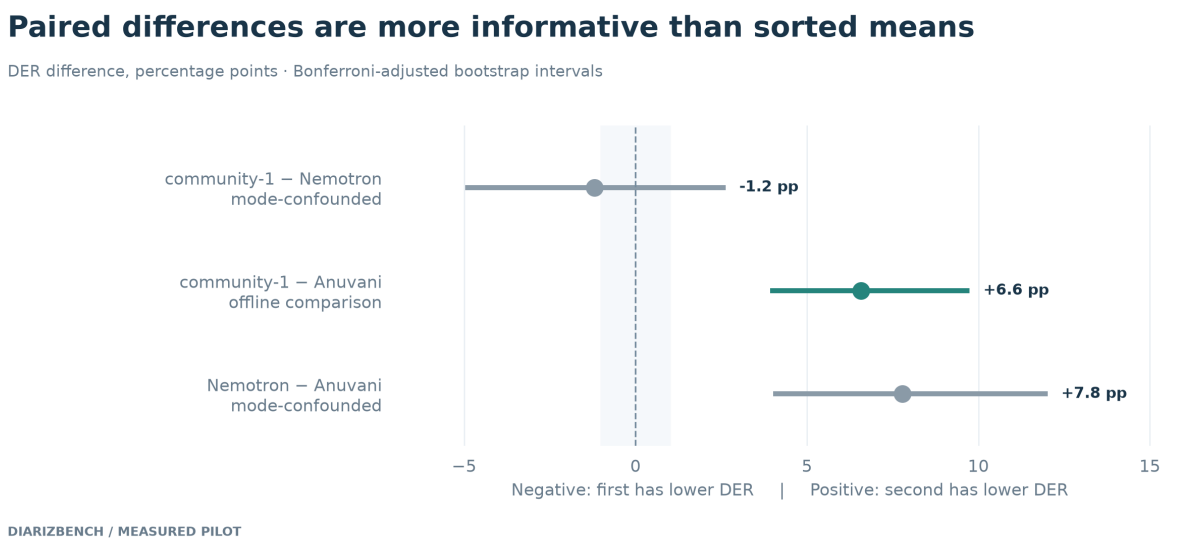


Figure 7. Paired DER differences with Bonferroni-adjusted bootstrap intervals. The community-1/Anuvani row is the matched offline comparison; the other two rows are mode-confounded. The shaded band marks plus/minus 1 pp.

The two metrics agree on one claim and disagree on another. Both put Anuvani ahead of community-1 in this sample under the stated test. On Nemotron, DER says the gap is large and significant (+7.8 pp) while JER says it is not established (+2.7 pp, CI spanning zero). JER equal-weights speakers, so the disagreement means Anuvani's DER and JER answer different weighted questions. These averages alone do not locate which speakers account for the difference; the JER gap is smaller and less certain. The defensible summary is:

> **Among the two offline systems, Anuvani has lower DER in this pilot. Its observed DER gap against the streaming Nemotron configuration is descriptive and cannot establish offline-model superiority. The community-1/Nemotron difference is not resolved; absence of significance does not prove equivalence. On JER, only the Anuvani–community-1 difference is established.**

This is the kind of statement a leaderboard should make it easy to produce and hard to overstate. With 24 recordings, the detectable difference is roughly 4 pp; anything smaller than that is a suggestion, not a result.

9. Three failure analyses

Each case is drawn from saved hypotheses and reference timestamps; `scripts/t10/analyze.py` reproduces the excerpts.

9.1 A brief speaker disappears (`BN-01`, Bengali near-field)

Reference speaker `SPEAKER_05` contributes **12.87 s, 4.6% of the meeting's speech**, in six short interjections across 168 seconds. All three systems drop the speaker (low-activity recall 0/1):

System	Hypothesis speakers	SPEAKER_05 recovered
community-1	4 of 8	no
Nemotron-3	6 of 8	no
Anuvani	6 of 8	no

The reference/hypothesis alignment suggests overlap with dominant turns and possible speaker merging. This is a timestamp-based interpretation, not a verified listening judgment: the speech time is largely still present, scored as confusion or miss, but the person has vanished from the transcript. This is the exact failure the low-activity diagnostic was designed to expose, and it is invisible in a DER table that shows 22.8–41.2% for the same file.

9.2 Three people talk at once (`TE-02`, Telugu in-the-wild)

At 97.2–97.7 s the reference has three simultaneous speakers:

- `Speaker_D` [92.1–97.7], `Speaker_A` [96.5–97.7], `New_speaker_A` [97.2–98.6]

The systems handle the same instant very differently. community-1 emits four labels in 0.6 s with 0.1–0.3 s slivers (over-segmentation). Nemotron emits three labels but splits one into two adjacent 90 ms fragments (boundary smear). Anuvani emits three local labels; their count alone does not establish correct identity or boundaries. No listening-based quality judgment was made. On the full recording, overlap-only DER on this file is 99.8 / 81.8 / 91.7%: even the cleanest output loses most of the overlap. Overlap handling, not clean single-speaker speech, is what separates these systems in practice.

9.3 A multilingual slice separates tested configurations (`TA-02` Tamil vs `TE-02` Telugu)

The source corpus describes code-mixing in its in-the-wild material. Without language-boundary annotations or a listening audit, these two examples support a multilingual comparison, not a causal claim about code-switching. The two Dravidian files tell different stories:

Recording	community-1	Nemotron-3	Anuvani	Failure mode
`TA-02` Tamil (3 speakers)	29.6	16.8	**5.7**	community-1: confusion 22.1 pp, collapses 3 to 2
`TE-02` Telugu (8 speakers)	38.6	42.0	38.4	all: missed + confusion, 20.4% overlap

Table 9. Two multilingual in-the-wild conversations. Code-switch boundaries were not annotated in this pilot.

On Tamil, community-1 is not missing speech (3.2 pp missed); it is confidently attributing speech to the wrong person (22.1 pp confusion) and losing a speaker entirely. Anuvani, fine-tuned on Indic conversation, is at 5.7%. On Telugu, the highest-overlap language in the pilot, everyone fails (38.4–42.0%) and the specialist's advantage nearly disappears. The pair is the strongest practical argument for inspecting Indic slices and a warning against attributing the gap specifically to code-mixing, model specialization, or microphone conditions without controlled comparisons.

10. Annotation audit: what the references disagree about

Six excerpts (2 Indic, 2 AMI, 2 VoxConverse) were audited structurally against a unified convention. This pass did not re-listen to audio; it characterises where the corpora's conventions diverge, which is the part that affects cross-corpus comparison.

Excerpt	Corpus	Region (s)	Speakers	Segments	Speech density	Overlap share	Segments < 0.5 s
HI-01	Indic	196.6	9	69	110.8%	16.1%	12
TE-02	Indic	300.0	8	163	122.9%	21.2%	20
AMI-02	AMI	2142.7	4	746	118.1%	20.5%	223
AMI-04	AMI	1505.6	4	242	68.1%	4.4%	79
VC-01	VoxConverse	300.0	13	49	100.8%	8.8%	5
VC-03	VoxConverse	300.0	8	35	91.6%	0.1%	1

Table 10. Structural audit of six references. Speech density is reference speaker-time divided by region duration; above 100% means overlap is being counted.

The audit establishes structural differences, not their cause or a measured DER penalty:

1. **Overlap coverage differs.** Observed overlap varies across these files, including very low values in several VoxConverse excerpts. This can reflect conversation content, annotation policy, or both. Without listening and re-annotation, low labelled overlap is not evidence of missing labels.
2. **Segment granularity differs.** AMI has many word-timed segments, while VoxConverse uses longer turns. Adjacent segments from the same speaker may merge into the same activity union; segment counts alone do not establish extra boundary error.
3. **Speech density is not speech occupancy.** The table divides speaker-time by region duration, so values above 100% are possible under overlap. It does not measure the fraction of silence. For example, AMI-02 has 118.1% density; AMI cannot be summarized as uniformly 68–88% speech.

No acoustic reference corrections were made. The six-file audit checks structure and timestamps; it does not establish that any reference is correct, defective, or internally acoustically consistent. Per-corpus reporting and exploratory cross-corpus aggregates are appropriate until a listening-based, blinded re-annotation study measures convention effects under a common handbook.

11. Did the research-inspired hypotheses hold?

#	Hypothesis from the reading	What the pilot observed	Verdict
H1	Count and low-activity diagnostics reveal failures DER hides (Beyond DER)	The community-1/Nemotron DER gap is not resolved; count accuracy differs by 20.8 pp and low-activity recall by 15.8 pp descriptively; no coverage superiority test was performed	**Supported**
H2	The Indic/code-mixed slice separates general from Indic-specialised systems (Indic DiarBench)	Anuvani leads Indic by 8.6–9.6 pp and Tamil by 9–17 pp — but also leads AMI and VoxConverse, so the advantage is not Indic-specific; Indic exposure is confounded	**Partially supported**
H3	Accuracy must be reported with measured speed, because there is a trade-off (SDBench)	No accuracy/speed trade-off appeared: Anuvani is both most accurate and lowest RTF. A load-time/throughput trade-off does exist and flips the ranking below ~11 min of audio	**Supported as a rule; predicted trade-off absent**
H4	Strict scoring is the honest default (DIHARD III)	Zero collar, overlap scored, unknown count produce a defensible ranking; 250 ms collar moves scores 2.8–3.6 pp without changing it; overlap-only DER is 60–68%	**Supported**

#	Hypothesis from the reading	What the pilot observed	Verdict
H5	Annotation conventions move DER materially (Anuvani)	Observed reference structure differs; content and convention effects remain confounded because no audio was re-annotated	**Partially supported**
H6	Error decomposition and crowded-recording analysis reveal failure modes (Benchmarking Diarization Models)	Components explain the ranking: Nemotron is 79% missed speech, community-1's Tamil failure is 75% confusion, the 8-speaker cap surfaces as a –5 count error	**Supported**
H7	Speaker changes and language changes are entangled in code-switched speech (DISPLACE)	Not tested: the pilot has no language-boundary annotations, so code-mixing enters only through corpus metadata	**Not tested**

Two of these verdicts cost something to write. H2 is weaker than the headline suggests: the tested specialist configuration has the lowest means across these samples. Base-model quality, training-corpus exposure, configuration, and adaptation effects remain confounded. H5 is the one we most wanted to quantify and could not, because quantifying convention effects requires re-annotating audio, not re-reading RTTM files.

12. Threats to validity

- **Sample size.** 24 recordings, two per Indic language. The detectable DER difference is about 4 pp; every per-language cell is exploratory. Anuvani's Indic margin is credible in direction, not in size.
- **Exposure.** Anuvani was fine-tuned on Indic DiarBench, one of the three pilot corpora. Its Indic result is an in-domain result. Its parent also lists AMI and VoxConverse as training corpora. community-1 has AMI/VoxConverse exposure; Nemotron's exposure is unknown. Exposure is printed on every score, and no cross-system claim ignores it.
- **Operating mode.** Nemotron ran streaming/low-latency, the other two offline. Its row is a streaming data point; its published offline Indic figure (18.38%) is 6.7 pp better than our 25.1%, on a different full-corpus protocol. It neither measures nor bounds a mode penalty here.
- **Corpus composition confound.** Indic near-field files have 5–9 speakers and 13–28% overlap; in-the-wild excerpts include 3–8 speakers and different overlap levels. The pilot's "near-field is harder" result (32.8 vs 13.0% for community-1) is confounded by speaker count and conversation content; it does not isolate a microphone effect.
- **Reference quality.** References were validated for internal consistency and speaker counts, not by listening. The audit is structural. A human listening pass is the obvious next step.
- **Scorer offset.** `pyannote.metrics` reads about 0.20 percentage points higher than NIST `md-eval` on Indic DiarBench according to Anuvani's card. Absolute values are not comparable to md-eval publications; relative comparisons here use one scorer consistently; a model-dependent offset is not ruled out.
- **Timing.** One GPU type and three repetitions. The timed callables differ in I/O and output post-processing scope; Nemotron returns frame logits. Setup/load can include dependency checks and Anuvani's self-test. Projected five-minute totals are not measured service latency.
- **Inference provenance.** Repository IDs and dependency constraints are recorded, but complete immutable weight revisions and environment locks were not consistently captured. Saved outputs support rescoring; a bit-for-bit future inference rerun is not guaranteed.
- **Excerpt truncation.** AMI is scored whole while the other corpora are excerpted, so file length is confounded with corpus. Per-corpus reporting contains the damage; it does not remove it.
- **No transcription-quality evaluation.** cpWER, tcpWER, WER/CER and related speaker-attributed ASR metrics were not run. The saved hypotheses contain activity labels, not recognized words.
- **Coverage and selection.** Three systems and three public corpora exclude telephone, healthcare, and many acoustic/language conditions. Unsuccessful integrations are disclosed; this is not a census of available systems or evidence that excluded models are worse.

- **Statistical interpretation.** Non-significant differences do not prove equivalence. Coverage gaps are descriptive, not tested superiority claims. Mixed-mode tests describe configurations; they cannot support a common offline/streaming model rank.

13. From pilot to production leaderboard

The pilot validates the instrument; the production design scales it. The full sketch is a separate deliverable; the operating rules that matter are these.

Aggregation. Eight domain × language-group cells (four domains × {English, Indic}), equal cell weights, with the conventional globally pooled number and every cell published alongside. The pilot approximates this with three corpus cells and six language slices, and says so.

UI references and implemented product. The supplied SignalBench HTML and MagicPath project informed the overview, results, comparison, audio inspector, datasets and evaluation-run sections. Their demo scores/settings and simulated audio were not used as evidence. The local HTML uses saved pilot results and three playable, attributed Indic DiarBench excerpts. It supports waveform seeking, speed controls, reference/model timelines, speaker-turn previews, local-file playback, run inspection and configuration export. Previews play the original mixture, not separated voices; configuration export does not launch inference. The PDF is a static sketch; the HTML is the functional prototype.

Table. Model and version, DER with interval, JER, count accuracy, RTF, and completion coverage in the main row. Filters for domain, language, speaker count, overlap, and operating mode. A model detail view exposes error components, counting bias, low-activity recall, hardware, settings, exposure, and one example failure per system, with an entry point to the audio inspector.

Operating mode is a filter, not a footnote. Streaming and offline results never share a rank. The pilot's Nemotron row is the cautionary example: without the mode label it would look like a worse model rather than a differently configured one.

Adding systems. Curated additions require a pinned checkpoint, a documented default configuration, an exposure declaration, and a complete run over the frozen manifest. Bring-your-own submissions are accepted only as a separate, clearly labelled track with self-reported settings and no rank until reproduced.

Re-scoring and versioning. References are versioned; a correction triggers a full re-score of all systems against the new reference version and publishes both the old and new numbers with the diff. Protocol versions are immutable, and every leaderboard row carries the protocol version that produced it. A system is never re-ranked silently.

Current scope and extension. Speaker-attributed transcription metrics (cpWER/tcpWER), language diarization, and live streaming latency were not measured. Section 14.4 proposes separately labelled tracks with their own references, settings, and timing protocol. These do not replace DER/JER.

13.1 Roadmap toward a speech evaluation lab (proposed)

A stronger research platform would diagnose individual stages as well as the complete product. These are proposed experiments, not capabilities evaluated by this pilot. They would share versioned datasets, consent/access records, immutable environments, held-out test splits, experiment tracking, automated checks, confidence intervals, and an evidence review before publication. A lab-quality process means controlled, reproducible experiments, not a promise of state-of-the-art scores.

Research track	Experiment and required reference	Outputs and decision
Speaker separation	Recover individual waveforms from an overlapping mixture using clean isolated sources or controlled mixtures. Compare mixture, separated stems and artifacts; test downstream diarization/ASR with and without separation.	Permutation-aligned SI-SDR and SI-SDR improvement, intelligibility/listening checks, downstream DER/tcpWER. Do isolated voices improve attribution without damaging words?

Research track	Experiment and required reference	Outputs and decision
Speaker similarity / verification	Extract embeddings from consented same/different-speaker pairs across microphones, sessions and languages. Keep identities disjoint between development and test; calibrate thresholds on development only.	Similarity distributions, equal error rate (EER), minimum detection cost (minDCF), false-accept/miss curves and calibration. Raw cosine similarity is not an identity probability.
Speaker clustering	Hold the speech segments and embeddings fixed; compare agglomerative, spectral and other clustering methods. Use oracle-count runs only as a separately labelled diagnostic.	DER/JER, count MAE, adjusted Rand index and homogeneity/completeness or V-measure. Inspect merging and fragmentation; purity alone can reward too many clusters.
VAD / SAD	Treat voice activity detection and speech activity detection as the speech-versus-non-speech stage; define filled pauses, backchannels and vocalizations consistently. Test noise, silence and overlapping speech.	Precision, recall, F1, missed-speech/false-alarm time and boundary errors. Fix downstream modules to isolate the effect of the detector and sweep thresholds on development data.
Controlled component studies	Freeze all but one stage: SAD, embeddings, clustering, overlap handling, separation, or ASR. Compare the same source conversations under matched modes and compute budgets.	Confidence intervals on end-to-end DER/tcpWER changes, RTF/memory/cost and failure cases. Separate a component improvement from a different dataset or implementation.

The listening room would eventually allow synchronized mixture versus separated-stem playback, consented speaker-pair listening, clustering merge/split inspection, and SAD overlays. At present it plays original excerpts and saved speaker turns only; those previews are not separated stems, computed similarity judgments, or a completed human annotation audit.

Method references: Asteroid source-separation toolkit, SpeechBrain verification metrics, and scikit-learn clustering evaluation.

13.2 Extending metrics, datasets, and deployment evidence (proposed)

With more time I would expand both **what is measured** and **where it is measured**, while retaining the existing DER/JER and speaker-coverage results as a separate diarization track.

1. **Add cpWER and tcpWER in a speaker-attributed transcription track.** cpWER (concatenated minimum-permutation word error rate) concatenates words by speaker and chooses the speaker permutation with the fewest substitutions, deletions, and insertions. tcpWER adds temporal constraints so words far apart in time cannot be matched merely because the text is similar. These answer whether the right words reached the right speaker stream; they cannot be computed from the pilot's RTTMs alone. Neither metric was run in this submission.
2. **Add complementary recognition and temporal diagnostics.** Report ordinary WER, and CER where language-specific tokenization makes it useful, to expose recognition errors independently of attribution. Consider DI-cpWER or ORC-WER to help distinguish recognition from speaker-assignment effects, plus speech-activity and overlap-detection precision/recall/F1 and boundary errors. Publish definitions, eligible samples, and uncertainty instead of combining unlike metrics into one opaque score. DER/JER, counting, low-activity recall, and RTF remain core outputs.
3. **Build the references these metrics need.** Human-check speaker-attributed transcripts; provide segment/word timings or explicitly document MeetEval's pseudo-word timing strategy. Freeze text normalization, Indic-script and code-mixed tokenization, punctuation/number policy, and tcpWER timing collar on development data. This timing collar is independent of DER's 0/250 ms collar. Use a pinned MeetEval version. For diarizer comparisons, hold the ASR model and word-to-turn attribution procedure fixed; show complete vendor ASR+diarization systems in a separate track.
4. **Evaluate more diverse datasets.** Add DIHARD III for diverse domains, ICSI for natural meetings, CALLHOME for telephone speech, DISPLACE for Indian multilingual/code-switched conversations, AliMeeting for Mandarin near/far-field meetings, and CHiME-8 DASR for distant multi-device speech. Verify access, licenses, held-out splits, speaker overlap with training, reference conventions, and transcript/timestamp availability before use. Give each corpus its own results; no new corpus is claimed as evaluated here. Grow toward the proposed 60-hour collection, with sample counts chosen using variance and power analysis rather than a fixed minimum that guarantees significance.
5. **Close the current fairness and reference gaps.** Run Nemotron on the same audio in documented offline and streaming configurations; isolate mode effects instead of borrowing a published score from

another protocol. Conduct a blinded listening audit and dual annotation/adjudication study, including brief speakers and overlap. Add language-boundary labels to test code-switching effects directly, and audit exact source/session exposure for all systems.

6. **Broaden deployment evidence.** Test CPU, GPU, and edge hardware over repeated sessions, measure cold/warm end-to-end time and cost per audio-hour, and add a live-input protocol for first-result delay, revision delay, and finalization latency. Add human-reviewed meeting-note attribution accuracy as a task-level outcome, separately from the acoustic and word metrics.

The practical sequence is: validate references and matching operating modes first; add a fixed-ASR cpWER/tcpWER pipeline next; then expand corpora and hardware. More metrics without reliable reference text, timestamp policy, and representative data would add apparent precision rather than evidence. Definitions and implementation reference: MeetEval, including its tcpWER protocol.

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14. **MeetEval: A Toolkit for Computation of Word Error Rates for Meeting Transcription Systems.** arXiv:2307.11394. https://github.com/fgnt/meeteval
15. **Additional dataset documentation:** https://dihardchallenge.github.io/dihard3/ ; https://groups.inf.ed.ac.uk/ami/icsi/ ; https://catalog.ldc.upenn.edu/LDC97S42 ; https://displace2024.github.io/ ; https://www.openslr.org/119/ ; https://www.chimechallenge.org/challenges/chime8/task1/

Appendix A. Reproducibility

Artefact	Location
Machine-readable manifest (24 rows, checksums, regions, exposure)	`harness/manifest.json`
Scoring harness + 25 tests	`harness/` (`.venv-harness/bin/pytest harness/tests`)
Reference extraction and audit	`scripts/extract_indic_references.py`, `scripts/audit_references.py`
Pilot driver (resumable, 72 units)	`kaggle/run_pilot.py`, `kaggle/diarizbench_models.py`
Kaggle notebook	`ayuwat/notebookb62243b3c5`
Merge, score, summarise	`scripts/t09/{merge_output,score,summarize}.py`
Failure analyses	`scripts/t10/analyze.py`
Statistics and figures	`scripts/t11/{statistics,figures}.py`
Uncertainty and sensitivity	`scripts/t12/uncertainty.py`
Document build	`scripts/t12/build_main.py`, `docs/main.md`

```
# rescore everything from saved hypotheses
.venv-harness/bin/python scripts/t09/score.py
.venv-harness/bin/python scripts/t11/statistics.py
.venv-harness/bin/python scripts/t12/uncertainty.py
.venv-harness/bin/python scripts/t11/figures.py
.venv-harness/bin/python scripts/t12/figures_doc.py
.venv-harness/bin/python scripts/t12/build_main.py --author "First Last"
```

Environment pins: `pyannote.audio>=4.0.1`, `torch<2.15`, `torchvision<0.27`, `numpy>=2.5,<2.6`, `numba>=0.67`, `transformers` (Nemotron), `anuvani_diar==1.0.0`. Audio is a private Kaggle dataset mounted at `/kaggle/input/datasets/ayuwat/diarizbench-pilot-audio/`. Full source recordings are not redistributed; the companion HTML embeds three attributed CC BY 4.0 excerpts totaling 47 seconds. Exact source intervals and license links are in `submission/AUDIO_SOURCES.md`.

Appendix B. Per-recording results

Recording	Language	Condition	Ref spk	c1 spk	Nemo spk	Anu spk	c1 DER	Nemo DER	Anu DER
AMI-01	English	SDM	4	4	4	4	25.2	33.2	18.5
AMI-02	English	SDM	4	4	4	4	23.9	31.8	19.7
AMI-03	English	SDM	4	4	4	5	21.8	22.7	18.0
AMI-04	English	SDM	4	2	4	3	26.1	39.9	19.8
AMI-05	English	SDM	4	3	4	3	29.0	34.2	20.0
AMI-06	English	SDM	4	4	4	5	10.0	16.0	8.8
BN-01	Bengali	near-field	8	4	6	6	41.2	28.5	22.8
BN-02	Bengali	in-the-wild	6	6	6	6	11.4	10.3	10.1
HI-01	Hindi	near-field	9	6	8	8	30.0	18.5	22.0
HI-02	Hindi	in-the-wild	5	4	4	4	9.1	10.2	4.6
KN-01	Kannada	near-field	7	5	4	4	24.7	36.9	16.7
KN-02	Kannada	in-the-wild	7	7	5	7	16.7	26.8	3.4
MR-01	Marathi	near-field	6	2	4	4	32.7	28.6	24.0
MR-02	Marathi	in-the-wild	5	4	5	4	10.2	7.4	6.2
TA-01	Tamil	near-field	5	3	3	4	20.6	16.9	9.8
TA-02	Tamil	in-the-wild	3	2	3	3	29.6	16.8	5.7
TE-01	Telugu	near-field	8	5	5	7	47.7	58.1	33.7
TE-02	Telugu	in-the-wild	8	7	7	9	38.6	42.0	38.4
VC-01	English	in-the-wild	13	13	8	13	14.4	28.2	13.3
VC-02	English	in-the-wild	8	7	8	8	4.8	2.1	2.7
VC-03	English	in-the-wild	8	8	8	8	5.4	2.1	4.5
VC-04	English	in-the-wild	8	7	8	7	8.0	3.6	4.5
VC-05	English	in-the-wild	6	6	6	5	5.6	1.9	2.1
VC-06	English	in-the-wild	6	5	6	5	2.2	1.1	2.1

Table B1. All 72 scored units. Speaker counts are in-region reference vs hypothesis. DER is zero collar, overlap scored.

Appendix C. Number provenance

Claim in this document	Source artefact
All DER/JER/count/low-activity/overlap/collar values per recording	`data/runs/results/<system>.json`, `<system>_collar0.25.json`
Pooling, bootstrap intervals, pairwise verdicts	`data/runs/results/statistics.json` (`scripts/t11/statistics.py`)
Count Wilson intervals, low-activity clustered intervals, 3×3 sensitivity grid, JER intervals	`data/runs/results/uncertainty.json` (`scripts/t12/uncertainty.py`)
Original figure sources	`scripts/t12/figures_doc.py`, `scripts/t11/figures.py`
Figures 1–7 (final visual design)	`scripts/t14/build_figures.py`; values from saved scores, statistics and uncertainty
RTF, load time, per-file ranges	`data/runs/timing/<system>/panel.json`, `panel_summary.json`
Failure-case timestamps and counts	`scripts/t10/analyze.py`, `t10_findings.md`
Run status, coverage, 72/72 units	`data/runs/state.json`, `summary.json`
Manifest, checksums, scoring regions, exposure	`harness/manifest.json`

Freeze statement. Measured scores remain unchanged. Sensitivity/uncertainty additions are saved in `uncertainty.json`; this revision changes presentation and proposed research, not measured results.

14. Reflections

14.1 Which design choice are you least sure about, and why?

The recording-equal headline aggregation. It gives a short conversation the same weight as a long meeting, which may not match a buyer's workload. I used it to keep one long meeting from dominating and published corpus-level and corpus-equal views alongside it. The observed ordering is stable across those summaries, but its margin changes. I would validate usage-based weights on held-out customer scenarios before defining a production headline. Zero collar is another trade-off, which is why the 250 ms sensitivity is published rather than substituted.

14.2 Which benchmark you read influenced you most, and how?

DIHARD III most influenced the methodology: it made me freeze scoring regions, collar, overlap handling and speaker-count policy before inference. **Beyond DER most influenced the product:** it motivated count accuracy and low-activity-speaker recall beside DER, so a missing participant cannot disappear inside a good average. Indic DiarBench motivated explicit multilingual slices; SDBench motivated measured speed with a stated workload. Each source changed a concrete choice rather than only appearing in the references.

14.3 What could make your leaderboard misleading, and how did you guard against it?

A single average can hide corpus imbalance, dropped speakers, training exposure, mismatched modes and uncertain differences. I publish per-corpus results, counting/JER/low-activity diagnostics, uncertainty, exposure and timing scope; offline and streaming have separate ranks. Failed or capacity-limited cases remain visible, and non-significance is not called equivalence. The audio inspector makes examples reviewable. Remaining gaps are the listening-based reference audit, exact session exposure and immutable inference provenance; displaying these gaps does not remove them.

14.4 What would you improve with more time?

First I would close the matched-mode, reference-quality and immutable-version gaps, then expand datasets and add cpWER/tcpWER with human-checked text, timing and a fixed ASR protocol. I would build separately scored **speaker separation, speaker similarity/verification, clustering, and VAD/SAD** tracks, using controlled ablations, held-out identities/conversations and task-appropriate metrics. The product would gain synchronized separated-stem and speaker-pair listening once those outputs exist. I would also measure hardware cost and live latency. Sections 13.1–13.2 specify the research roadmap; none of those additional model evaluations is claimed as completed here.